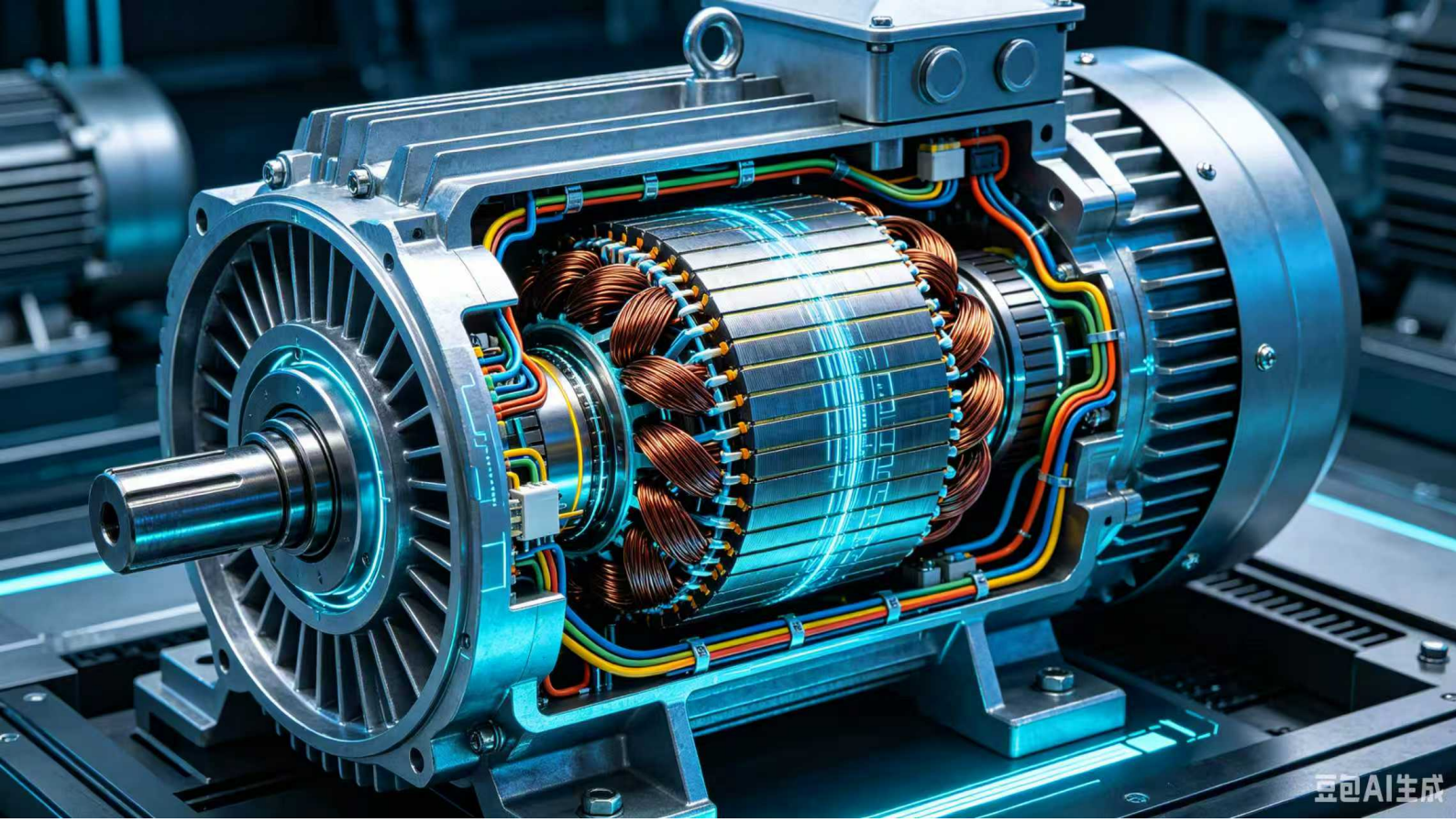


9. Magnetic Field of a Current

> Experiments are the only means of knowledge at our disposal.
> The rest is poetry, imagination.
> – *Attributed to Max Planck*

9.1 Magnetic Field due to a Current

A steady current produces a magnetic field that is constant in time, and the produced field is therefore called a **steady magnetic field** as well. Unlike electrostatics, the current element $id\mathbf{l}$ that produces the magnetic field is a vector, whereas the charge element dq is a scalar. The magnetic field produced by such a current element at a position P located $\mathbf{r}$ from the element is:

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{id\mathbf{l} \times \mathbf{r}}{r^3}$$

This is called the **Biot–Savart law**. An isolated steady current element cannot exist, so this law is tested through its integral around a closed steady circuit. For any steady current distribution, we can first divide the current into small segments and apply this law to obtain the magnetic field contribution from each small segment, and then obtain the total magnetic field by integration.

The direction of the magnetic field produced by a current follows the **right-hand grip rule**: Grasp the current-carrying wire with your right hand, with your extended thumb pointing in the

direction of the current. Your fingers then naturally curl around in the direction of the magnetic field lines. For a current-length element $id\mathbf{l}$, the magnetic field direction is given by the cross product $id\mathbf{l}\times\mathbf{r}$. For a curved wire, apply the rule to each element, which gives the direction of that element's contribution; only for a straight wire do the contributions add up to circles around the wire.

Example 9.1: A circular ring carrying current i.

We shall calculate the magnetic field on the z axis, at a distance z from the center of the ring (Figure 9.1figure.9.1), and assume that the radius of the ring is a. The current circulates counterclockwise as seen from the positive z axis. Taking a small piece of current element from θ to $\theta + d\theta$ contributes:

$$d\mathbf{l} = \hat{\theta} a\, d\theta$$

therefore:

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{i\hat{\theta}a\, d\theta \times (-a\hat{r} + z\hat{z})}{(a^2+z^2)^{3/2}} = \frac{\mu_0}{4\pi}\frac{ia\, d\theta(a\hat{z} + z\hat{r})}{(a^2+z^2)^{3/2}}$$

Integrating around the ring, the terms containing $\hat{r}$ cancel, and we have:

$$\mathbf{B} = \hat{z}\frac{\mu_0}{4\pi} ia^2 \int_0^{2\pi} \frac{d\theta}{(a^2+z^2)^{3/2}} = \hat{z}\frac{\mu_0}{2}\frac{ia^2}{(a^2+z^2)^{3/2}} \tag{9.1}$$

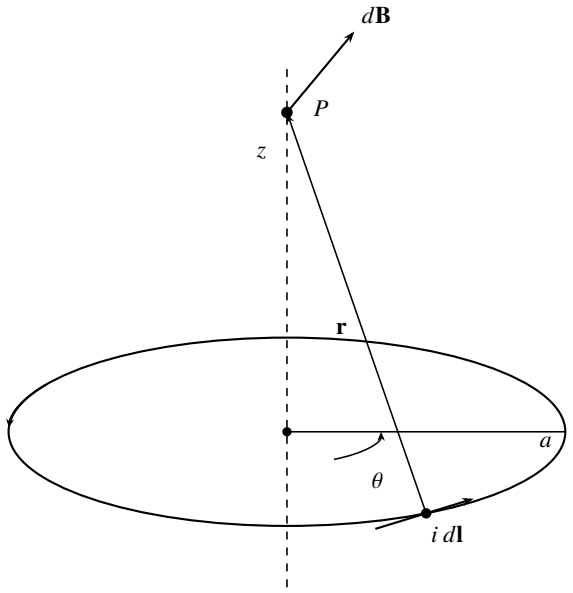


Figure 9.1: Magnetic field on the axis of a circular current loop of radius a. A current element $i\, d\mathbf{l}$ produces $d\mathbf{B}$ at point P. By symmetry, the radial components cancel, leaving only the z-component.

Example 9.2: Infinitely long straight wire.

We take the straight wire along the z axis (Figure 9.2figure.9.2), carrying current i. The point of interest is located a distance x from the wire in the $\hat{x}$ direction. A small piece of current from z to $z + dz$ contributes:

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{i\hat{z}dz \times (x\hat{x} - z\hat{z})}{(x^2 + z^2)^{3/2}}$$

Thus,

$$\mathbf{B} = \hat{y}\frac{\mu_0}{4\pi} \int_{-\infty}^{\infty} \frac{ixdz}{(z^2 + x^2)^{3/2}} = \hat{y}\frac{\mu_0}{2\pi}\frac{i}{x} \tag{9.2}$$

For the last step, you can substitute $z = x \tan\theta$. This is the standard result for an infinite wire.

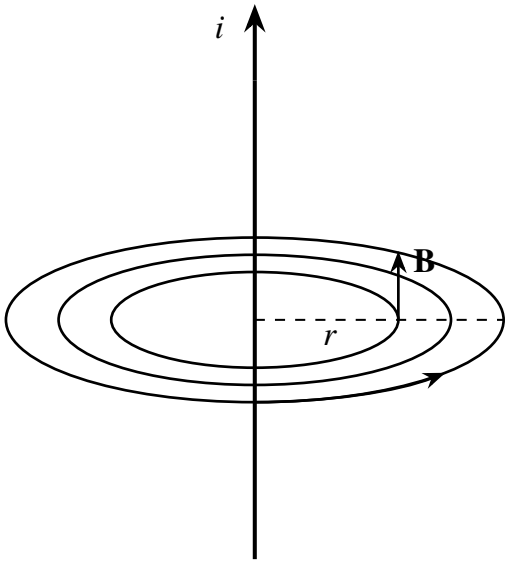


Figure 9.2: Magnetic field of an infinitely long straight wire computed using the Biot–Savart law. The current i flows upwards, and the field lines form concentric circles around the wire in the sense given by the right-hand rule, with magnitude $B = \mu_0 i/(2\pi x)$.

Combining with the expression of the Ampère force, we can now calculate the force between two parallel wires. Let the wires be long enough to be treated as infinitely long, with currents i_A and i_B and separation d. The magnetic field produced by the first wire at every point along the second wire is then:

$$B = \frac{\mu_0}{2\pi}\frac{i_A}{d}$$

The direction is perpendicular to the plane formed by both wires. Thus, the force per length on the second wire is:

$$\frac{F}{L} = \frac{\mu_0}{2\pi}\frac{i_A i_B}{d}$$

and the direction would be either pointing to the first wire, or away from it, depending on the relative directions of i_A and i_B. You can verify that if both currents are in the same direction, then the force is attractive; and the force is repulsive if the current is in opposite directions. Since $\mu_0 = 4\pi \times 10^{-7}$ N/A^2, we immediately see that the force per meter between two parallel wires separated by 1 m with 1 A current is 2×10^{-7} N per meter of length. Between 1948 and 2019 the Ampère was defined

by this force, which fixed $\mu_0 = 4\pi \times 10^{-7}$ N/A^2 exactly; since 2019 the ampere is defined by fixing the elementary charge, and μ_0, the vacuum permeability, is a measured quantity that agrees with $4\pi \times 10^{-7}$ N/A^2 to a few parts in 10^{10}.

9.2 Ampère's Law

Consider the field of a long straight wire, $B = \mu_0 i/(2\pi r)$ around the wire. Along any path, $\mathbf{B} \cdot d\mathbf{l} = (\mu_0 i/2\pi)\, d\phi$, where ϕ is the angle around the wire. For a closed path that goes once around the wire the angle changes by 2π, so $\oint \mathbf{B} \cdot d\mathbf{l} = \pm\mu_0 i$, with the plus sign when the path goes around the current in the right-hand sense; for a closed path that does not go around the wire the angle returns to its starting value, and the circulation is zero; and a path that goes around the wire m times, counted with sign, gives $m\mu_0 i$. This is the so-called *circulation* of the magnetic field generated by an infinitely long straight wire.

Ampère's law generalizes this result. It states:

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \sum_{\text{enc}} i \tag{9.3}$$

where the summation means the algebraic sum of all current enclosed by the path C. It is convention to assign a positive sign for the current, if the integration direction of the path and the direction of the current obeys right-handed rule; and the current has a negative sign if the situation is the opposite. If the current is not discretely distributed, we can write a more general form of it as:

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{j} \cdot d\mathbf{S}$$

The integral on the right hand side takes place on a surface bounded by C. The surface is oriented by the right-hand rule: with the fingers curled along the direction of C, the thumb gives the side toward which $d\mathbf{S}$ points.

Example 9.3: Two current-carrying wires and two paths.

The sign with which each current enters Ampère's law is fixed by the sense in which the path encircles it. In the figure, two long straight wires perpendicular to the page carry the currents i_1 (out of the page, marked $\odot$) and i_2 (into the page, marked $\otimes$), and two closed paths are drawn (Figure 9.3figure.9.3). The path C_1 encircles both wires, and is traversed counter-clockwise as seen from the reader; the right-hand rule then counts a current coming out of the page as positive and a current going into it as negative, so that

$$\oint_{C_1} \mathbf{B} \cdot d\mathbf{l} = \mu_0(i_1 - i_2).$$

If the two currents have the same magnitude, this circulation vanishes, even though the field $\mathbf{B}$ itself does not vanish anywhere on the path. The path C_2 encircles the first wire only, and is traversed in the opposite sense, so that

$$\oint_{C_2} \mathbf{B} \cdot d\mathbf{l} = -\mu_0 i_1.$$

Reversing either the sense of the traversal or the direction of the current flips the corresponding sign.

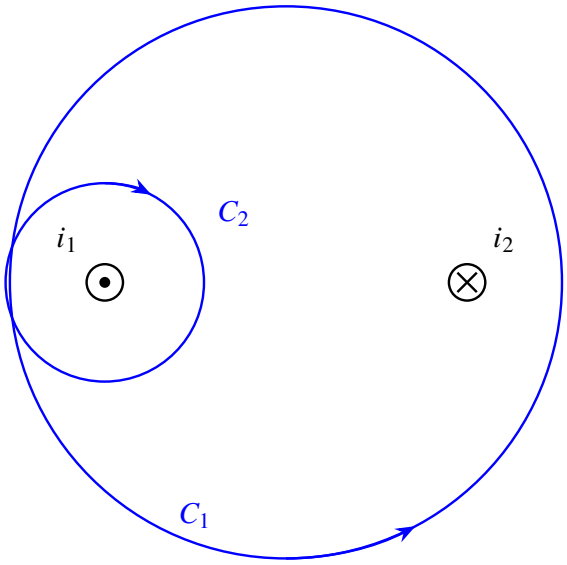


Figure 9.3: Two long straight wires perpendicular to the page, carrying the currents i_1 out of the page ($\odot$) and i_2 into the page ($\otimes$). The path C_1 encircles both wires and the path C_2 encircles only the first; the arrow on each path gives the sense of traversal used in Ampère's law.

Similar to Gauss' law, we can apply Ampère's law to obtain the magnetic field under certain highly symmetrical situations.

Symmetry arguments need one change from Chapter 2. The field **B** is built from a cross product, so when the currents are reflected in a mirror, the field at the mirror-image point has its component perpendicular to the mirror unchanged and its components parallel to the mirror reversed — the opposite of **E**. When in doubt, read the direction off the Biot–Savart law, as in Example 9.2.

Example 9.4: Infinitely long straight wire again.

We use the same setup as before. By symmetry, we immediately know that the magnetic field **B** must be azimuthal, and that its magnitude depends on nothing else but the distance between the point of interest and the wire. Therefore we take the loop path as a circle of radius r around the wire (Figure 9.4figure.9.4). Therefore we have:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \int B dl = 2\pi r B = \mu_0 i$$

Thus, $B = \frac{\mu_0 i}{2\pi r}$.

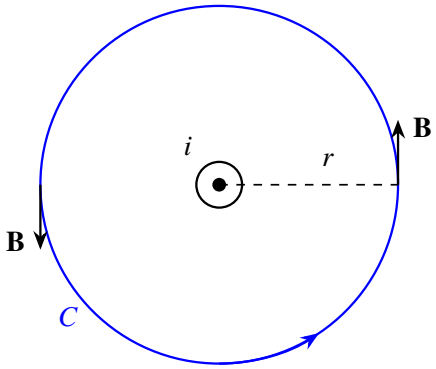


Figure 9.4: The same infinite straight wire treated with Ampère's law, seen end on: the current i comes out of the page. A circular Amperian loop of radius r gives $\oint \mathbf{B} \cdot d\mathbf{l} = B \cdot 2\pi r = \mu_0 i$.

Field inside a cylindrical wire. If the current is uniformly distributed over the cross section of a wire of radius R, then for a point at distance $r < R$ from the axis, Ampère's law (with $i_{\text{enc}} = i \cdot \pi r^2 / \pi R^2$) gives:

$$B = \frac{\mu_0 i}{2\pi R^2} r.$$

Thus the field increases linearly with r inside the wire and falls off as $1/r$ outside.

Example 9.5: A Slab of Moving Charge.

We shall assume that the slab is infinite in two directions, of thickness w and carrier density n, and that the charge drifts with speed v. Employing the definition of current density, we immediately see that $j = nev$. Based on symmetry analysis, we see that the magnetic field would be parallel to the plate and perpendicular to the moving direction. We use a rectangular loop path with one edge in the direction of the magnetic field (Figure 9.5figure.9.5). Ampère's law then reads:

$$\oint \mathbf{B} \cdot d\mathbf{l} = 2Ba = \mu_0 \int \mathbf{j} \cdot d\mathbf{S} = \mu_0 nevaw$$

Thus

$$B = \frac{\mu_0}{2} nevw$$

outside the slab, on either side. Here $e > 0$ is the magnitude of the carrier charge, and $\mathbf{j}$ points along the carriers' velocity for positive carriers and against it for negative ones. Inside the slab, a path whose long sides lie a distance $|y| < w/2$ above and below the mid-plane encloses the thickness $2|y|$, so there $B = \mu_0 nev|y|$.

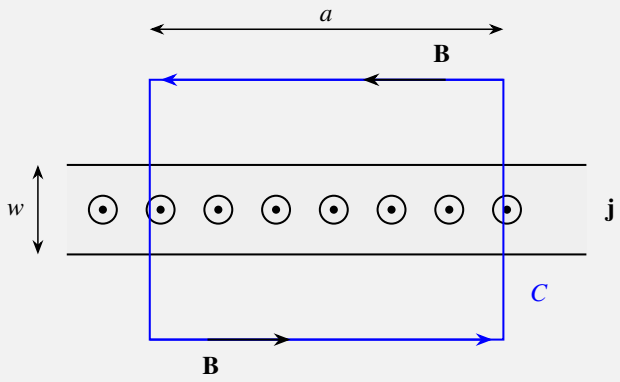


Figure 9.5: A slab of moving charge of thickness w, seen edge on, carrying the current density $j = nev$ out of the page. The rectangular Amperian loop C straddles the slab, with its two long sides of length a parallel to $\mathbf{B}$; Ampère's law gives $B = \mu_0 J_s/2$ outside the slab, the same magnitude on both sides, in opposite directions, where the surface current density is $J_s = nevw$.

Defining the surface current density J_s (current per unit width perpendicular to the flow), this can be written compactly as $B = \mu_0 J_s/2$, independent of distance from the sheet outside it—analogous to $E = \sigma/(2\epsilon_0)$ for an infinite sheet of charge.

Example 9.6: Infinitely long solenoid

Again, we use a rectangular loop path with one edge parallel to the axis (Figure 9.6figure.9.6). We treat the winding as a sheet of current flowing only around the axis. Outside the coil the field then has no azimuthal component, since no current flows along the axis, and no radial component, since its flux through a closed cylinder around the coil is zero (Sec. 9.3) and nothing depends on position along the axis. A rectangular path lying entirely outside the coil encloses no current, so the axial component outside has the same value at every distance from the axis; since it must vanish far from the coil, it is zero everywhere outside. Inside, the field is parallel to the axis, and Ampère's law for the path of Figure 9.6figure.9.6 becomes:

$$\oint \mathbf{B} \cdot d\mathbf{l} = Ba = \mu_0 \sum i = \mu_0 nai$$

where n is the number of turns per length. Thus:

$$B = \mu_0 ni \tag{9.4}$$

This is the known result.

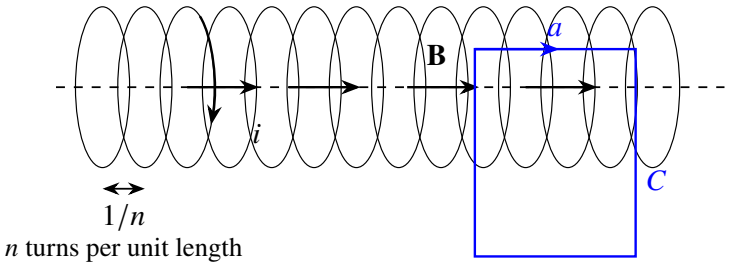


Figure 9.6: An infinitely long solenoid with n turns per unit length, seen from the side. The magnetic field inside is uniform, $B = \mu_0 ni$ along the axis, while outside $B = 0$ for a winding that carries current only around the axis, as an ideal solenoid does. The rectangular Amperian loop C has one side of length a inside the coil and one side outside it.

Example 9.7: Torus

In this case we consider a torus with a circular cross section, wound uniformly and tightly with N turns of wire, and choose a circular loop path of radius r (Figure 9.7figure.9.7). If the loop is within the tube of the torus, then:

$$\oint \mathbf{B} \cdot d\mathbf{l} = 2\pi r B = \mu_0 \sum i = \mu_0 Ni$$

and we have:

$$B = \frac{\mu_0}{2\pi} \frac{Ni}{r}$$

Otherwise, both outside the torus and in its hole, $B = 0$.

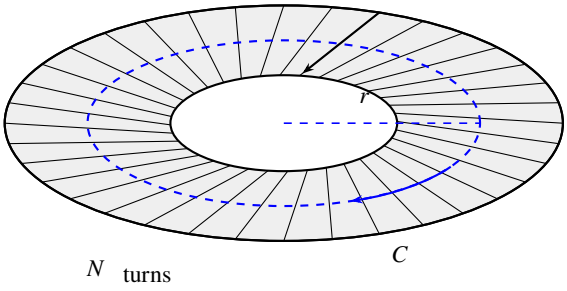


Figure 9.7: A toroidal coil (torus) with N turns, wound uniformly and tightly. The dashed circle is the Amperian loop of radius r inside the tube; Ampère's law applied to it yields $B = \mu_0 Ni/(2\pi r)$, while $B = 0$ outside the torus and in its hole for such an ideal winding.

Although the paths in our previous examples all appear to be in a plane, there is no such constraint in Ampère's law. Moreover, in the general form, there is not even a constraint on which surface the integration should take (there are an infinite number of surfaces bounded by a path). Under a steady current situation, the continuity equation automatically ensures that the integral yields exactly the same result for all surfaces. However, it is not the case when the current is varying in time. This is what led Maxwell to add the displacement current, and to what we now call Maxwell's equations.

In addition, applying Stokes' theorem — for a vector field $\mathbf{A}$ and a surface S bounded by a closed path C, oriented by the right-hand rule, $\oint_C \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$ — we obtain the differential form of Ampère's law. Applying it to the left-hand side,

$$\int_S (\nabla \times \mathbf{B} - \mu_0 \mathbf{j}) \cdot d\mathbf{S} = 0$$

for every surface S bounded by C, and only because the integral vanishes for every such surface does the integrand vanish:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$$

9.3 Gauss' Law for Magnetism

No magnetic monopole (magnetic charge) has ever been observed, and we take it that none exists, which means

$$\oint \mathbf{B} \cdot d\mathbf{S} = 0$$

or

$$\nabla \cdot \mathbf{B} = 0$$

Nevertheless, this equation can also be derived from the Biot–Savart law, assuming the magnetic field is generated by a steady current. Replacing the current element $i\, d\mathbf{l}$ by $\mathbf{j}(\mathbf{r}')\, d^3\mathbf{r}'$, the Biot–Savart law for a steady current distribution reads

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{j}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3\mathbf{r}'$$

Its divergence reads:

$$\nabla \cdot \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \cdot \frac{\mathbf{j}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3\mathbf{r}'$$

Note that $\mathbf{r}'$ can be viewed as constant for the operator ∇, since it only operates on $\mathbf{r}$; using the identity $\nabla \cdot (\mathbf{C} \times \mathbf{D}) = (\nabla \times \mathbf{C}) \cdot \mathbf{D} - \mathbf{C} \cdot (\nabla \times \mathbf{D})$, what we need is both

$$\nabla_{\mathbf{r}} \times \mathbf{j}(\mathbf{r}') = 0$$

and

$$\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3} = -\nabla\frac{1}{|\mathbf{r}-\mathbf{r}'|}$$

is a gradient, and the curl of any gradient is zero; because $1/|\mathbf{r}-\mathbf{r}'|$ is integrable at $\mathbf{r}' = \mathbf{r}$, this holds there too when the integral is differentiated. The divergence of a gradient is its Laplacian, which need not vanish: for this field it is concentrated at $\mathbf{r}' = \mathbf{r}$, the flux through any small sphere around that point being 4π, which is why $\nabla \cdot \mathbf{E}$ is not zero inside a charge. Therefore $\nabla \cdot \mathbf{B} = 0$.

9.4 Exercises

1. Three straight leads carry a current I in the directions shown in Figure 9.8figure.9.8. The middle lead has length l. The two outer leads are semi-infinite; each is joined to one end of the middle lead, lies on the side of it away from P, and makes an angle of 45° with the line of the middle lead. The foot of the perpendicular dropped from P onto the middle lead is a distance a from its left end ($0 < a < l$), and P is a distance b from the lead. Calculate the magnitude and direction of the magnetic field at point P.

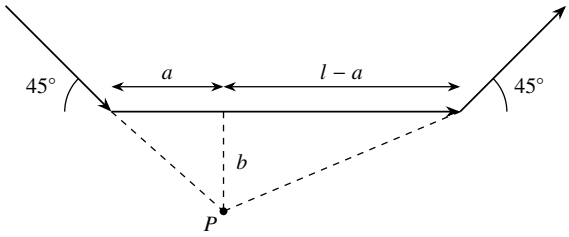


Figure 9.8: Three straight leads; the middle lead has length l, and the two outer leads are semi-infinite, each lying on the side of the middle lead away from P and making 45° with its line.

2. A sphere with radius R has a uniform charge distribution on its surface, carrying a total charge Q. The center of the sphere is at the origin, and the sphere is rotating around the z axis with angular velocity ω in a counterclockwise fashion, as seen from the positive z axis. Calculate the magnitude of the magnetic field at point $(0, 0, z)$ for both $|z| < R$ and $|z| > R$.

3. A long straight coaxial cable consists of a conducting cylinder with radius a, and a conducting tube with inner radius b and outer radius c ($a < b < c$). Currents of same magnitude I but opposite directions are passing through the system (Figure 9.9figure.9.9). We assume that the length of the cable is infinite, and that the current density is uniform within each conductor. Calculate the magnitude of the magnetic field B at radius r for:

   (a) $r < a$;

   (b) $a < r < b$;

   (c) $b < r < c$;

   (d) $r > c$.

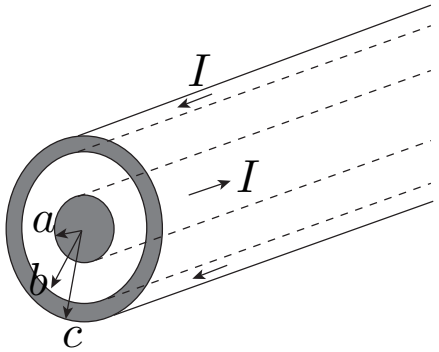


Figure 9.9: Two opposite currents.

4. Consider two infinitely long cylinders of radius R whose parallel axes are a distance $2a$ apart ($R > a$). In the shaded non-overlapping area two opposite currents are passing through, where the current density is uniform with magnitude j, and no current flows in the overlap. Calculate the magnitude and direction of the magnetic field at any point inside the overlapping area.

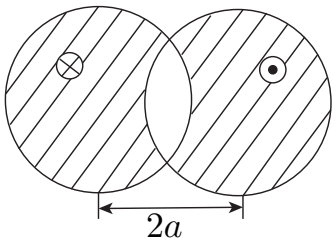


Figure 9.10: Two overlapping cylinders of radius R, with their centers separated by $2a$; opposite currents flow in the shaded non-overlapping regions.

5. Two infinitely large thin inter-penetrating boards are assembled as shown in Figure 9.11figure.9.11, such that the three dimensional space is divided into four regions. A current is flowing towards the $+y$ direction on the board in the xy plane, and a current is flowing towards the $+z$ direction on the board in the xz plane. We assume the boards are infinitely large and thin, so that the current in each is a surface current, with surface current density J_s. Calculate the magnitude of the magnetic field in each region of the space.

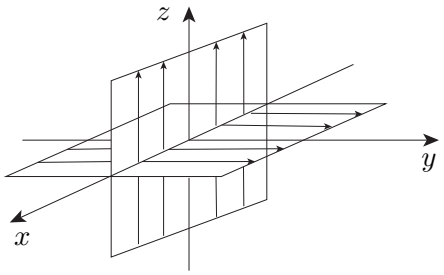


Figure 9.11: Two infinite thin conducting boards.

6. Two long, straight, parallel wires, each of mass λ per unit length, hang side by side from a common horizontal axis parallel to them: each wire is held along its length by light insulating threads of length L attached to the axis. When the wires carry equal currents I in opposite directions they repel each other, and in equilibrium the threads of each wire make an angle θ with the vertical. Treating the wires as infinitely long, find I in terms of λ, L, θ, μ_0 and the gravitational acceleration g.

7. Two long straight wires cross the plane of the page at right angles, at the points A and B. The wire through A carries a current i_1 out of the page, and the wire through B carries a current i_2 into the page, with $i_1 > 0$ and $i_2 > 0$. Evaluate $\oint \mathbf{B} \cdot d\mathbf{l}$ along each of the following closed paths, both of which lie in the plane of the page.

   (a) A path that goes around A twice, counterclockwise as seen by the reader, without going around B.

(b) A figure-eight path whose two loops lie side by side and meet at a single point, where the path crosses itself rather than merely touching itself: one loop goes around A counterclockwise as seen by the reader, and the other goes around B.

8. A long solenoid of radius R is wound from a single wire as a tight, single-layer helix with n turns per unit length, and the wire carries a steady current i. Because the wire advances along the axis as it winds, it crosses every cross section of the solenoid once, so the winding carries the current i along the axis as well as around it. Model the winding as a thin cylindrical sheet of radius R that carries a current ni per unit length around the axis and, in addition, a total current i along the axis, spread uniformly around its circumference. Treat the solenoid as infinitely long and ignore its leads. Let r be the distance from the axis.

   (a) Find the magnitudes of the axial and of the azimuthal components of the magnetic field inside ($r < R$) and outside ($r > R$) the solenoid.

   (b) Find the ratio of the magnitude of the field just outside the winding to its magnitude inside.

9. An ideal toroidal coil of N turns carries a steady current I. Its tube has a circular cross section of radius a, whose center runs around a circle of radius R ($R > a$) in the xy plane, centered at the origin. The coil is wound so that the magnetic field inside the tube circulates counterclockwise as seen from the positive z axis. A thin, straight test wire is threaded through the winding along the x axis, from $x = R - a$ to $x = R + a$, and carries a current i in the $+x$ direction. Neglect the field produced by the test wire itself.

   (a) Find the magnetic force on this segment of wire (not on its leads).

   (b) Let B_0 be the magnitude of the field on the central circle of the tube. Show that the magnitude of the force is larger than $2aiB_0$, the force that a uniform field of magnitude B_0 would exert.

10. Each of the following time-independent vector fields, in which k is a nonzero constant, is proposed as the magnetic field in some region of space. For each, decide whether it can be a magnetic field. If it can, find the current density $\mathbf{j}$ that must flow in that region; if it cannot, find its outward flux through the surface of the cube $0 \le x \le s$, $0 \le y \le s$, $0 \le z \le s$, where $s > 0$.

    (a) $\mathbf{B} = k\,(x\hat{x} - y\hat{y})$.

    (b) $\mathbf{B} = k\,(x\hat{x} + y\hat{y})$.

    (c) $\mathbf{B} = kx\hat{y}$.